

PROJECT TITLE

Railway Track Fall Detection System Using Region Based Convolutional Neural Networks for People Safety

PROBLEM STATEMENT

Railway tracks develop cracks, breaks, and other defects over time due to heavy loads, weather changes, and continuous use. If these defects are not detected early, they can cause major accidents such as derailments. Traditional track inspection is done manually by workers, which is slow, time-consuming, and cannot cover long tracks regularly.

Therefore, there is a need for an automatic track fault detection system that can continuously monitor tracks, identify cracks or defects early, and alert railway authorities to prevent accidents and ensure safe train operation.

ABSTRACT

This project aims to create an automatic system that can continuously monitor railway tracks and detect faults at an early stage. The system uses technology such as sensors or image processing to scan the tracks and identify any unusual patterns, cracks, or damage. When a fault is detected, the system immediately sends an alert to railway authorities so that repairs can be done quickly.

By detecting track faults early, this system helps improve railway safety, reduce accidents, and ensure smooth and reliable train operations.

EXISTING METHODS

1. Manual Track Inspection

Railway staff physically walk along tracks to visually inspect for cracks or damage.

- **Highly labor-intensive and slow**
- **Prone to human error**
- **Impractical for long-distance routes**

2. Ultrasonic Testing

Sound waves are transmitted into rails to detect internal cracks based on signal reflections.

- **High detection accuracy** for internal defects
- Requires **skilled operators and special equipment**
- **Periodic testing only**, not continuous monitoring

3. Camera-Based Visual Inspection

Fixed or moving cameras capture images or videos of railway tracks for analysis.

- **Still depends on manual review** of footage
- **Time-consuming** for large video data

- Performance drops during **long monitoring hours**

4. Vibration and Strain Sensor Monitoring

Sensors measure rail stress and vibration patterns to infer abnormalities.

- Enables **real-time condition monitoring**
- **Sensitive to environmental and load variations**
- **Limited to indirect fault detection**

5. Track Recording Vehicles

Dedicated vehicles equipped with advanced sensors inspect tracks while in motion.

- Provides **highly detailed and reliable data**
- **Very high cost and operational complexity**
- **Used occasionally**, not suitable for daily monitoring

DISADVANTAGES

- Manual inspection is slow and can miss faults.
- Ultrasonic testing needs skilled workers and is not continuous.
- CCTV monitoring depends on humans and may miss cracks.
- Vibration sensors give false alarms and need maintenance.

PROPOSED METHODS

The proposed system employs Region-Based Convolutional Neural Networks (R-CNN) for automated railway track fault detection.

Steps Involved

- Capture railway track images/videos using cameras.
- Preprocess data (resizing, normalization).
- Use R-CNN to generate region proposals.
- Extract features using CNN.
- Classify regions as normal track or faulty track.
- Localize faults such as cracks, breaks, or missing tracks.
- Send alerts for maintenance action.

Key Idea:

R-CNN not only detects whether a fault exists but also pinpoints the exact defective region, improving response time and safety.

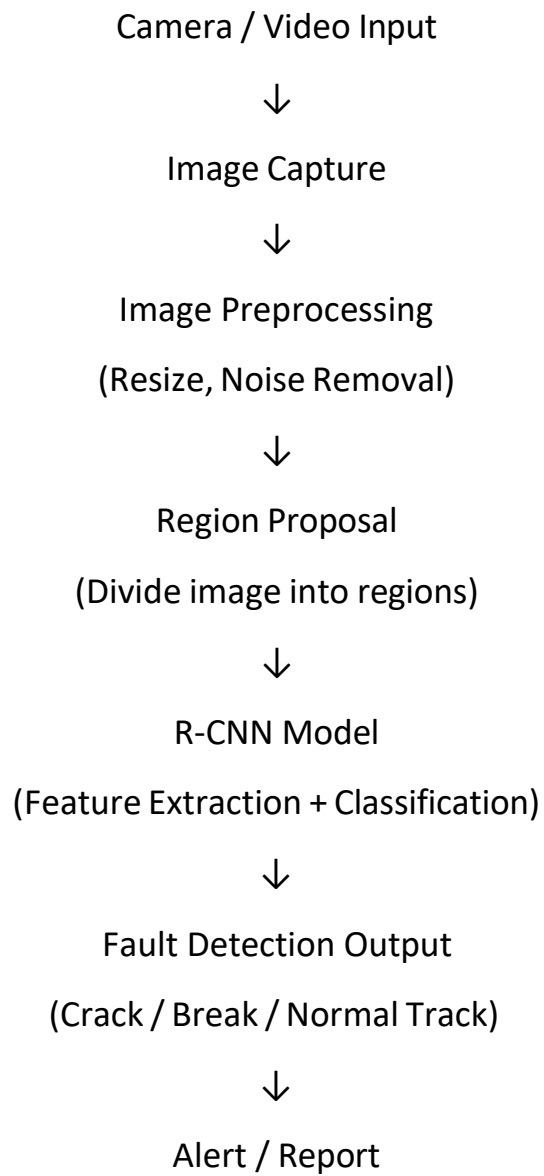
ADVANTAGES

- Accurate fault detection with precise location identification
- Early detection enables timely maintenance and prevents accidents
- Reduces manual inspection effort, human error, and operational cost
- Supports automated monitoring using real-time track images or videos
- Scalable and suitable for long railway routes, improving overall safety

ARCHITECTURE/BLOCK DIAGRAM

- **Camera Input** – Captures live video of railway tracks.
- **Frame Extraction** – Converts video into individual frames for analysis.
- **Pre-processing** – Resizes, normalizes, and cleans frames for better accuracy.
- **Region Proposal Network (RPN)** – Identifies areas that may contain falls or objects.
- **Feature Extraction (CNN)** – Extracts important patterns like edges and shapes from regions.
- **Classification & Localization** – Determines whether a fall, object, or safe track is present and locates it.
- **Decision Module** – Confirms detected events and decides if an alert is needed.
- **Alert System** – Sends notifications, alarms, or messages to authorities.
- **Monitoring Dashboard** – Displays real-time alerts and visual information for control-room staff.

BLOCK DIAGRAM:



REQUIREMENTS

Software Requirements

- Operating System: Windows / Linux
- Programming Language: Python
- Libraries: OpenCV, NumPy
- Framework: TensorFlow or PyTorch
- IDE: VS Code

Hardware Requirements

- HD Camera
- Computer or Laptop
- Minimum 8 GB RAM
- Sufficient storage
- GPU (optional, recommended)